

1. Coin Change Problem:

The Coin Change Problem is an optimization problem in which we are given a set of coin denominations and a target amount. The goal is to find the minimum number of coins required to make the given amount.

ALGORITHM:

1. Read the number of coin denominations.
2. Read the values of the coins.
3. Read the target amount.
4. Create a table to store the minimum number of coins required for each amount.
5. Initialize the table with a large value (infinity), except for amount 0, which requires 0 coins.
6. For each amount, check every available coin.
7. If a coin can be used, update the minimum number of coins.
8. Continue until the target amount is reached.
9. Display the minimum number of coins required.

Time Complexity:

Case	Time Complexity
Best Case	$O(n \times A)$
Average Case	$O(n \times A)$
Worst Case	$O(n \times A)$

Space Complexity:

Space Complexity = $O(n \times A)$

OUTPUT:

```
Enter the number of coin denominations: 3
Enter the coin denominations: 1
2
5
Enter the amount: 10
Minimum number of coins required: 2

...Program finished with exit code 0
Press ENTER to exit console.
```

CONCLUSION:

The Coin Change Problem was successfully solved using the dynamic programming approach. The algorithm finds the minimum number of coins required to form the given target amount efficiently.